

# VINAY PADALA

AWS Engineer | SRE | MLOps Engineer | 10 YOY | AZURE | EKS |  
CI&CD | Airflow

✉ [vinay.padala07@gmail.com](mailto:vinay.padala07@gmail.com)

☎ +1 470-403-8085

in <https://linkedin.com/in/vinay-padala-452571128/>

📍 Atlanta, GA

## SUMMARY

- **AWS Certified Solution Architect & SRE Engineer** with 9+ years of experience in designing, implementing, and maintaining scalable, secure, and resilient cloud infrastructure across AWS and Azure.
- **Cloud Expertise:** Hands-on experience with AWS (EC2, EKS, ECS, S3, RDS, Lambda, EMR, OpenSearch, VPC) and Azure (Azure DevOps, VMs, Blob Storage, Azure Functions, Azure Monitor, Azure SQL).
- **Infrastructure as Code (IaC):** Proficient in Terraform, CloudFormation, and ARM templates for provisioning and managing infrastructure in AWS and Azure.
- **Configuration Management:** Experience with configuration management and infrastructure as code tools including Chef, Puppet, Ansible, and AWS CDK.
- **CI/CD & Automation:** Expertise in building and maintaining CI/CD pipelines using Jenkins, Bamboo, Azure DevOps, GitHub Actions, Tekton, and ArgoCD to enable seamless and automated deployments.
- **Containerization & Orchestration:** Skilled in Docker and Kubernetes (EKS, AKS), with experience in Helm, Karpenter, and autoscaling for high-availability deployments.
- **Monitoring & Logging:** Implemented observability solutions using AWS CloudWatch, Splunk, Azure Monitor, Prometheus, and Grafana for system performance and availability.
- **Networking:** AWS Networking Specialist with hands-on experience designing and implementing secure, scalable network architectures using VPC, Direct Connect, Transit Gateway, and Load Balancers. Proficient in optimizing performance and enforcing security through CloudFront, security groups, and perimeter security solutions.
- **Apache Airflow:** Designed and orchestrated complex data pipelines using Airflow for scheduling and monitoring data workflows across AWS and Azure.
- **Airflow Pipeline Design and Maintenance:** Pioneered the implementation and ongoing maintenance of data pipelines in Airflow, streamlining the automation of data processing tasks for improved operational efficiency.
- **MLOps:** Experience supporting ML pipelines with Airflow, S3, EKS, SageMaker (AWS), and Azure Machine Learning, enabling scalable and repeatable ML model operations.
- **Data Engineering:** Worked on ETL pipelines using AWS Glue, EMR, Lambda, and Azure Data Factory, integrating large-scale structured and unstructured data from various sources.
- **Cost Optimization & Governance:** Implemented automated cost tracking, resource tagging, and right-sizing strategies to reduce cloud spend and enforce compliance.
- **Security & Compliance:** Applied best practices in IAM, VPC security groups, Azure RBAC, and infrastructure scanning to maintain compliance and secure infrastructure.
- **DevOps in Healthcare Domain:** Delivered solutions for HIPAA-compliant infrastructure in healthcare applications, ensuring data privacy and operational excellence.
- **Agile & Collaboration:** Strong background in Agile environments, collaborating with cross-functional teams to ensure smooth delivery cycles and rapid feedback loops.
- **HIPAA:** Health data privacy, secure transmission, breach notification policies.
- **GDPR:** Personal data handling, consent management, right to erasure (right to be forgotten), data breach response.
- **Certifications:** AWS Certified Solutions Architect - Associate

## TECHNICAL SKILLS

- **AWS Cloud :** AWS(EC2, ECS, EKS, EMR, S3, LoadBalancer, Lambda, ASG, EFS, CDN, RDS, ElasticCache, EMR, Glue, OpenSearch, RedShift, CloudWatch, VPC, Route53, IAM, WAF, SageMaker, SNS, SQS, LightSail, BeanStalk, Apache Airflow)
- **Azure Cloud :** Azure(VM, AKS, ACI, Azure Functions, Blob Storage, Files, CDN, SQL, Log Analytics, VNET, Application Gateway, NSG)
- **Containerization :** Docker, Kubernetes
- **CI/CD Tools :** Jenkins, Bamboo, Tekton, GitLab Actions, ArgoCD

- **VCS** : GIT, BitBucket
- **Configuration Management** : Terraform, CloudFormation, Ansible, Chef, Ansible, Puppet
- **Opertaion System** : Linux, Windows, Unix
- **Scripting Language** : Shell, Python
- **Monitoring Tools** : DataDog, Splunk, CloudWatch, Grafana
- **DevSecOps Tools** : Sonar Scanner, Veracode, Checkmarx, Wiz.io

## WORK EXPERIENCE

Carelon Global  
Solutions(Client: Elevance  
Health)  
Hyderabad, India  
Nov 2019 - Mar 2025

### Senior Cloud Engineer

- Architected and managed public and private cloud infrastructure using AWS services, including **VPC, EC2, S3, CloudFront, EFS, RDS, Route 53, ECS, EKS, ECR, AWS Glue, OpenSearch, Gateways, Auto Scaling Groups, and Lambda**, aligning solutions with business requirements.
- Designed and implemented fully automated **CI/CD pipelines** using **Tekton** and **Bamboo** for seamless code deployments into AWS infrastructure, ensuring high reliability and efficiency in software delivery.
- Reduced infrastructure costs by 30% by optimizing AWS resource allocation, leveraging **Karpenter** for dynamic EKS node provisioning, and implementing **Auto Scaling Groups**.
- **Accelerated CI/CD pipeline execution by 50%**, reducing deployment time from **40 minutes to under 20 minutes** by optimizing Tekton and Bamboo workflows .
- Designed and implemented fully automated CI/CD pipelines using Tekton and Bamboo ,ensuring 99.9% deployment reliability and **zero-downtime releases**.
- **Migrated from AWS ECS to AWS EKS**, improving workload orchestration and **scalability by 3x** while reducing operational overhead.
- **Refactored application deployment pipelines** for Kubernetes-native workflows in **EKS** and **Elastic Beanstalk**, ensuring seamless integration with existing CI/CD processes.
- **Optimized Kubernetes infrastructure with Karpenter**, enabling **dynamic scaling**, **reducing idle compute costs by 25%**, and improving cluster efficiency.
- **Deployed and managed OpenSearch as a backend data store**, ensuring **sub-50ms query response times** and high availability for mission-critical applications.
- Implemented SRE best practices, including **error budgets, SLOs, automated incident response**, and **self-healing architectures**, improving **MTTR (Mean Time to Resolution) by 40%**.
- **Enhanced observability** and incident response using **CloudWatch, Splunk, and DataDog Dashboards**, proactively detecting and resolving 95% of issues before impacting users.
- **Automated AWS infrastructure provisioning using Terraform and CloudFormation**, ensuring 100% consistency across environments and reducing manual errors.
- **Played a key role in AWS Landing Zone (NLZ) implementation**, improving **security, governance, and compliance** adherence across multiple environments.
- **Developed self-healing architectures** using Kubernetes health probes, AWS Lambda, and automated failover mechanisms, reducing downtime and improving availability.
- Conducted post-mortem analysis for incidents, identifying root causes and implementing corrective actions to prevent recurrence, reducing incident frequency by 35%.
- **Implemented chaos engineering principles** to proactively test failure scenarios and improve system stability.
- **Orchestrated data processing workflows using AWS EMR Serverless and Apache Airflow**, improving data processing efficiency by 60%.
- **Developed Python automation scripts to streamline operational tasks**, reducing manual intervention by 40%.
- **Managed GitHub repositories**, enforcing branching strategies, permissions, and CI/CD integrations, **ensuring seamless collaboration and code**

stability.

- **Led zero-downtime deployments**, implementing canary releases, blue-green deployments, and rollback automation, reducing deployment failures by 90%.
- **Promoted Agile and Scrum methodologies**, ensuring faster feature delivery and reducing release cycles from 4 weeks to 2 week.
- **Collaborated with cross-functional teams** to enhance system reliability, resulting in a **30% reduction in downtime and improved user satisfaction**.
- Collaborated with cross-functional teams to integrate **Azure DevOps** pipelines for specific healthcare compliance modules.
- Managed small-scale **Azure Kubernetes Service (AKS)** workloads and deployed Proof of Concept workloads on **Azure Functions** and **Blob Storage**.
- Used **Azure Monitor** and **Log Analytics** for telemetry and comparison during hybrid cloud assessments.
- Collaborated with data teams to support **data lake ingestion** pipelines using **AWS Glue, Airflow**, and **S3 lifecycle policies**.
- Created dashboards for **capacity forecasting** and cost tracking using **QuickSight** and **Grafana**.
- Mentored 3 junior engineers on AWS and DevOps tools, increasing team efficiency and cloud maturity.
- Maintained documentation, playbooks, and recovery strategies for critical services hosted across **AWS** and **Azure**.
- Experience in infrastructure automation and configuration management using **Chef**
- Proficient in managing system configurations and deployments with **Puppet**.
- Hands-on experience automating IT workflows and provisioning with **Ansible**.
- Skilled in defining cloud infrastructure using **AWS Cloud Development Kit (CDK)** in code.
- Implemented **Wiz.io CSPM** for AWS/Azure, **enabling agentless cloud visibility, vulnerability scanning, misconfiguration detection**, and real-time risk-based alerting to improve security and compliance posture.
- Implemented **secure infrastructure solutions** ensuring **compliance** with **HIPAA** for patient data protection.
- Supported **GDPR** compliance initiatives by designing **secure access controls, audit logging**, and **data encryption mechanisms**.

Cloud Automation  
Solutions(Client: TierPoint &  
Cisco)  
Hyderabad, India  
*Jan 2016 - Oct 2019*

## Cloud Engineer

- **Managed and optimized cloud environments** (DEV, QA, and Production), **ensuring seamless deployments and high availability**.
- **Designed and implemented end-to-end build and release processes**, streamlining deployment for both **on-premises and cloud-based applications**.
- **Configured and maintained CI/CD pipelines using Jenkins**, automating application builds, testing, and deployments.
- **Provisioned and managed AWS RDS databases**, implementing **automated backups to S3** for data protection and recovery.
- Administered critical AWS services including **IAM, EC2, RDS, Route 53, S3, SES**, and **SNS**, ensuring security, scalability, and fault tolerance.
- **Automated infrastructure provisioning and management** using **Cisco Process Orchestrator** and **StackStorm**, enabling self-service deployment workflows.
- **Integrated Git with Jenkins**, ensuring **version control best practices**, code quality enforcement with **SonarQube**, and security scanning using **Veracode**.
- Streamlined deployment processes for both on-premises and cloud-based applications, significantly cutting down release times and deployment errors.
- Developed **custom shell and Python scripts** for routine operational tasks such as log rotation, backup automation, and service health checks.
- Led **incident response efforts**, root cause analysis, and implementation of preventive measures, reducing critical incidents by 30%.
- Monitored application and infrastructure health using **AWS CloudWatch**, creating

alarms and dashboards for proactive alerting and capacity planning.

- Supported **DevSecOps practices** by incorporating security checks into pipelines and ensuring role-based access and least privilege policies across AWS accounts.
- Configured **AWS Direct Connect** to establish low-latency, dedicated network connections between on-premises infrastructure and AWS, improving data transfer speeds and reducing costs.
- Deployed and managed **AWS Transit Gateway** to enable scalable and simplified connectivity between multiple VPCs and on-prem networks.
- Integrated **Amazon CloudFront** as a CDN layer to accelerate global content delivery and reduce latency for end users.
- Created and managed **security groups** and **network ACLs** to enforce granular access control at the instance and subnet level.

## PROJECTS

Hyderabad, India  
*Jun 2023 - Dec 2024*

### Migration from ECS to EKS, Carelon Global Solutions

- **Transitioned containerized applications** from Amazon ECS to Amazon EKS, significantly enhancing workload orchestration, scalability, and portability.
- **Designed and implemented CI/CD pipelines** using **Tekton** for Kubernetes-native build and deploy workflows, enabling faster and more reliable releases.
- Utilized **ArgoCD** for GitOps-based multi-environment deployments, ensuring consistency, rollback capability, and improved release traceability.
- **Created Helm charts** for reusable Kubernetes manifests, simplifying application deployment and configuration management across environments.
- **Optimized Kubernetes cluster autoscaling** with **Karpenter**, reducing cost and improving resource utilization by dynamically provisioning nodes.
- Implemented **Service Mesh (Istio)** to enhance traffic control, observability, and security between microservices.
- **Set up monitoring and alerting** using **Prometheus** and **Grafana**, improving visibility into cluster health and application metrics.
- Automated infrastructure provisioning using Terraform, ensuring consistency across dev, test, and prod EKS environments.
- Migrated data pipelines and background jobs to run in EKS with **Kubernetes CronJobs** and **AWS Lambda**, achieving better scheduling and reliability.
- Conducted knowledge transfer and documentation sessions to upskill internal teams on EKS architecture, best practices, and deployment patterns.

Hyderabad, India  
*Jan 2021 - Jun 2022*

### Migration of Apache Solr to OpenSearch and RDS, Carelon Global Solutions

- **Led the end-to-end migration** of large-scale search indexes from **Apache Solr** to **AWS OpenSearch**, enhancing scalability, high availability, and real-time search capabilities.
- Migrated relational data from legacy systems to **Amazon RDS (MySQL/PostgreSQL)**, improving database reliability and performance.
- Utilized **AWS Glue crawlers** and custom **ETL scripts** for data discovery, schema inference, and efficient data transformation during migration.
- **Orchestrated batch and streaming data pipelines** using **AWS EMR** and **Apache Spark**, ensuring high-throughput processing and minimal data latency.
- Created **custom analyzers and tokenizers** in OpenSearch to support advanced search use cases such as autocomplete, fuzzy search, and multilingual queries.
- Enhanced search response times by over **60%** post-migration by tuning OpenSearch configurations and optimizing query execution.
- Designed **fault-tolerant and restartable ETL jobs** in Glue and EMR, ensuring minimal data loss in the event of job failures.
- Integrated **OpenSearch Dashboards** for visualizing search analytics and monitoring cluster health metrics.
- Documented and trained internal stakeholders on managing OpenSearch clusters, best practices for index optimization, and RDS performance tuning.

Hyderabad, India  
Dec 2019 - Jun 2020

## Migration from On-Premises to AWS, Carelon Global Solutions

- Designed and deployed **scalable and secure AWS infrastructure using CloudFormation templates (CFT)**, ensuring consistency and repeatability across multiple environments (DEV, QA, UAT, PROD).
- Migrated monolithic and microservices-based applications to **Amazon ECS**, leveraging **Fargate** and **EC2 launch types** based on workload characteristics and cost-efficiency.
- Implemented **Auto Scaling Groups (ASG)** and **Application Load Balancers (ALB)** to handle dynamic traffic patterns and ensure high availability.
- Integrated **AWS WAF** with CloudFront and ALB to protect against OWASP Top 10 vulnerabilities and DDoS attacks.
- Used Amazon **Route 53** for domain management, routing policies, and global traffic distribution.
- Integrated **Akamai CDN** for low-latency content delivery, enabling global acceleration of static and dynamic content with edge caching.
- Used **Elastic File System (EFS)** for shared storage across ECS tasks and **Amazon S3** for scalable object storage and static site hosting.
- Conducted **performance testing and cost analysis**, optimizing ECS task definitions, container images, and auto-scaling rules to reduce **compute cost by 20%**.
- Coordinated **disaster recovery (DR)** strategies with multi-AZ deployment architecture and automated failover mechanisms.

Hyderabad, India  
Aug 2016 - Aug 2019

## Cloud & Infrastructure Provisioning, Cloud Automation Solutions

- **Designed and maintained** scalable cloud infrastructure supporting high-availability applications across **multiple environments**.
- **Monitored and optimized** system performance using **CloudWatch, Prometheus, and Grafana**, improving reliability and uptime.
- **Implemented security** best practices across **infrastructure**, including **encryption**, least privilege access, and auditing.
- Collaborated with development teams to support **microservices** architecture and containerized applications using **Docker**.
- Led migration of **legacy systems** to **cloud-native solutions**, reducing operational overhead and **increasing deployment speed**.
- Wrote custom scripts in **Python** and **Bash** to automate routine tasks and infrastructure validations.
- Participated in **disaster recovery planning** and setup, ensuring business continuity through automated backups and failover strategies.

## CERTIFICATIONS

Jan 2022

### AWS Certified Solutions Architect Associate by Amazon Web Services

## EDUCATION

New Horizon College of  
Engineering  
Bangalore, India  
Jun 2012 - May 2016

### Bachelor of Education in Computer Science